environmentally friendly operations. The efficiency of the power source directly influenced range, payload, and flight duration.

- *Payload module*: The payload module was a mission-specific component designed to carry sensors, cameras, or scientific instruments. These payloads enabled applications in surveillance, reconnaissance, cartography, meteorology, and environmental monitoring. Modular payload systems allowed quick reconfiguration, transforming the same UAV for different missions, thus enhancing operational flexibility and cost-effectiveness.
- *Ground control station (GCS)*: The ground control station served as the command center for UAV operation. It provided an interface for flight control, mission planning, and data visualization. Equipped with telemetry displays, communication consoles, and live video feeds, the GCS allowed operators to monitor flight performance, adjust navigation parameters, and coordinate multi-UAV operations in real time.

Table 3.2 summarizes the essential components and functional architecture of second-generation drones, highlighting the integration of structural, electronic, and communication systems. Each component contributed to improved stability, precision, and mission versatility, marking the transition from manually operated UAVs to semi-autonomous, GPS-guided aerial platforms. The synergy between the flight controller, navigation modules, and telemetry systems established the technological framework for modern unmanned aerial systems (UAS).

Table 3.2 Components and functions of second-generation drones.⏎

Component	Description/function	Significance